the axioms that define a complex vector space (in Section 1.9.1), we can see that complex functions satisfy all of them:

1. The sum of any two functions is a function.

2. The addition of functions is commutative.

3. The addition of functions is associative.

4. There is a unique *zero function* such that when you add it to any function, you get the same function back.

5. Given any function $\psi(x)$, there is a unique function $-\psi(x)$ such that $\psi(x) + (-\psi(x)) = 0$.

6. Multiplying a function by any complex number gives a function and is linear.

7. The distributive property holds, which means that

$$z[\psi(x) + \phi(x)] = z\psi(x) + z\phi(x)$$
$$[z + w]\psi(x) = z\psi(x) + w\psi(x),$$

where z and w are complex numbers.

All of this implies that we can identify the functions $\psi(x)$ with the ket-vectors $|\Psi\rangle$ in an abstract vector space. Not surprisingly, we can also define bra vectors. The bra vector $\langle\Psi|$ corresponding to the ket $|\Psi\rangle$ is identified with the complex conjugate function $\psi^*(x)$.

To use this idea effectively, we'll need to generalize some of the items in our mathematical tool kit. In earlier lectures, the labels that identified wave functions were members of some finite discrete set—for example, the eigenvalues of some observable. But now the independent variable is